

Quant Risk Mini-Project : Portfolio VaR + Stress Testing

Objective

You will build a small risk engine for a 3-stock equity portfolio using market data from Yahoo Finance and produce:

1. Portfolio Value-at-Risk (VaR) using multiple methods
2. Stress tests / scenario analysis

This is deliberately time-boxed. We care more about clear thinking & correctness than perfect polish.

Setup & Data

Use 3 liquid US stocks (you may pick your own, but a good default is):

- AAPL (Apple)
- GOOGL (Alphabet)
- MSFT (Microsoft)

Data Source

Pull daily adjusted close prices from Yahoo Finance

Portfolio Definition

Assume a \$1,000,000 portfolio with weights:

- AAPL: 40%
- GOOGL: 35%
- MSFT: 25%

(If you prefer, you may implement weights as inputs/config.)

Part A — VaR Calculation

Compute 1-day VaR at 95% and 99% confidence using at least two of the following (three is a bonus):

Method 1: Historical Simulation VaR

- VaR from empirical portfolio return distribution
- VaR is the appropriate return quantile translated into \$ loss

Method 2: Parametric (Variance–Covariance) VaR

Assume returns are normal:

- portfolio volatility from covariance matrix
- $\text{VaR} = z * \sigma * \text{portfolio_value}$

(Bonus: show also Expected Shortfall / CVaR under normal assumption.)

Method 3: Monte Carlo VaR

Simulate N scenarios for 1-day returns:

- simplest: multivariate normal using mean vector + covariance matrix
- compute simulated portfolio P&L distribution
- VaR from simulated quantile

Part B — Stress Testing & Scenarios

Implement at least 2 stress scenarios and compute the resulting portfolio P&L.

You can choose from:

Scenario 1: “2008-style equity shock”

Apply an instantaneous shock to prices (or returns) for 1 day:

- AAPL -8%, GOOGL -7%, MSFT -6%
Compute portfolio loss in dollars.

Scenario 2: “Tech drawdown”

- All three stocks -10% same day
Compute portfolio loss.

Scenario 3: “Correlation spike”

Keep individual volatilities the same, but increase pairwise correlations to 0.85 and recompute Parametric VaR (95/99).

Scenario 4 (Bonus): Rolling VaR breach backtest

Compute rolling historical VaR(95) over a window (e.g., 250d) and count exceedances.

- expected exceedance rate $\approx 5\%$
- show number of breaches and comment

Engineering Requirements (applies to all parts)

- Code runs end-to-end: `python main.py` (or a notebook with “Run All”)
- Clear separation of concerns:
 - `data.py` (fetch/clean)
 - `risk.py` (VaR methods)
 - `scenarios.py` (stress logic)
 - `main.py` (wires it together)
(You can do fewer files if you keep it clean.)
- Functions should be testable (inputs/outputs), not just inline scripts
- Reasonable error handling

Evaluation Rubric (how you will be assessed)

Understanding (50%)

- Correct VaR interpretation (loss quantile, confidence meaning)
- Correct portfolio aggregation (weights, covariance)
- Correct scenario P&L logic
- Basic understanding of assumptions and limitations

Engineering Quality (40%)

- Clean, readable, modular code
- Reproducibility
- Output is easy to interpret

Submission Format

Zip folder with:

- source code
- requirements.txt
- README with run instructions & relevant plots / stats
- (optional) notebook + exported HTML/PDF